

MAPPINGS - an Introductory Guide

(version 1.0.3)

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Purpose of this document

This document is an introduction to MAPPINGS. It describes what MAPPINGS does, how to install it, the input and output parameters, with examples, including how to run a simple HII region model and generating isolated single atom models.

Who should read this? And how?

The document is aimed at beginning users and as a refresher for people who have not used MAPPINGS for some years, or who have only used much older versions. It is intended to be read in a linear mode, from start to finish.

What is MAPPINGS and what does it do?

MAPPINGS is photoionisation modelling code written in FORTRAN. It allows one to compute nebular emission spectra from the far UV ($\sim 300 \text{ \AA}$) to the far IR ($> 600 \text{ }\mu$). It calculates emission line fluxes for recombination lines, intercombination lines and collisionally excited lines for all elements from hydrogen to zinc, where these exceed a (presettable) minimum flux. MAPPINGS began in 1976 as a five-level-atom solver. It has gone through major upgrades since then and is currently at version 5.1

It can be used to model HII regions, Planetary Nebulae, AGN emission regions and shockwaves in the interstellar medium.

It is unique among major photoionisation modelling codes in handling emission lines from shocks. It is also (as far as we know) the only code capable of working with kappa non-equilibrium electron energies. Further, it incorporates element abundance scaling based on the abundance patterns in old stars, rather than uniform linear scaling, which we know to be wrong (Nicholls, Sutherland et al, 2017, [ADS 2017MNRAS.466.4403N](#), especially Figure 1).

It uses the latest and best available atomic data, drawing on the Chianti8 database where possible (Del Zanna, Dere et al., 2015, [ADS 2015A&A...582A..56D](#)).

What this document covers (and does not cover)

As MAPPINGS is a complex and sophisticated application, writing a complete user manual is a daunting task. We propose to incorporate the current document as part of a wiki which will expand piecewise as we cover the full range of MAPPINGS capabilities. In the meantime, this document provides a guide to starting with MAPPINGS. It explores only modelling HII region models. Discussion of modelling AGN spectra and shock emission lines will be added later.

Downloading MAPPINGS

The latest public version of the MAPPINGS source code (5.1.13) is available from <https://mappings.anu.edu.au/code/>. Initially, download "Everything including stellar/agn atmospheres" (217MB).

What to know before you start

To install and run MAPPINGS, you should be familiar with running commands from the terminal. At some point, you may need to edit pre-defined command scripts to set input parameters and to point to the data directories, depending where the data files are located with respect to the working area. The standard work area is the /lab directory. Download MV_atmos.zip Stellar atmospheres (192.8MB). Unzip to create the atmos directory and move the atmos directory into the lab area.

Installing MAPPINGS

These directions are intended for Mac users but are readily adaptable to Linux and other Unix versions. They apply to OSX version 10.9 (Mavericks) onwards. In what follows, we use the `Courier New` typeface for terminal commands and file names.

For maximum computational efficiency you will need a FORTRAN compiler. MAPPINGS will compile with a variety of FORTRAN compilers, but probably the most useful is `gfortran`. This runs on Linux (various flavours) and Mac OSX platforms.

An alternative is to install the `f2c` application which can be found in the /src directory in the MAPPINGS download. This converts the MAPPINGS code to compilable C, for compilation using the Apple or other C compilers. This results in a somewhat slower compiled application. Also note that if you need to compile MAPPINGS this way, you will need to use the correct `make` file (in the /src directory). Rename `Makefile.f2c` to replace the default `Makefile` in the /src directory.

Before you can run the standard `Makefile` on OSX, it is necessary to install Apple's Xcode on the Mac, and the command line tools. These are available via the Appstore. This is a large download but is necessary for installing appropriate compilers.

To install `gfortran` on Mac OSX, you will next need to install Macports. See <https://www.macports.org>. When the Macports framework is installed, the next step is to install the GNU compiler collection from the command line. This is done with the command: `sudo port install gcc8`. This installs both the GNU C compiler (version 8) and the `gfortran` compiler (version 8).

At present we recommend you use the latest `gfortran` version, currently `gfortran8`. If you are using an older version of Mac OSX (for example, 10.9 Mavericks, because you have an old Mac computer), `gfortran` version 4.9 is the better choice. Also note that in this case you will need a legacy Macports install. Refer to the Macports website for details. To check the `gfortran` compiler is installed and available, use the following command in the terminal: `gfortran --version`. If this does not confirm the version, you will need to seek assistance.

When Xcode, the command line tools, Macports and the GNU compilers are installed, unzip the mappings download into a new directory. The MAPPINGS makefile is found in the /src directory.

Install in the standard way from the /src directory using `make install`. Full installation instructions are included in the `README.txt` file. This is a comprehensive description of the installation and you should read it carefully before commencing. It is included as Appendix 1 in this document.

Open the readme file or go to the appendix below and follow the instructions. MAPPINGS can be run from the /lab directory, but the best approach, as described in the readme file, is to make a universal install which can be run from any location using the command `map51 .`

MAPPINGS data inputs

There are several types data input for MAPPINGS: physical model parameters (pressure, density, ionisation parameter etc.), atomic data, dust depletion data, abundance scaling data, standard abundances, and excitation spectra.

The main data are contained in extensive libraries in the sub-directories in the /lab directory. The HIIGrid306 directory contains additional libraries of abundances and pre-calculated synthetic cluster spectra derived from the Starburst99 spectral synthesis code. The input data are described in more detail below.

Ways to run models

You can run MAPPINGS models in three ways: entering input data an entry at a time at the input prompts; pasting a single model file at the first MAPPINGS input prompt; or running a multi-model shell script, for example to create $\log(Q)$ /metallicity grids for strong line ratios.

You can also use Mappings to run single atom models at specified temperatures and densities, but while useful, these do not correspond to physical nebulae.

Running a single model

In the terminal, change to the /lab directory. Run the command `./map51`. You will be presented with a series of prompts as MAPPINGS requests the data it needs to run a model. These relate to abundance data, type of model, physical parameters (e.g., pressure, starting temperature, ionisation parameter etc). We will now describe the main prompts and responses. For the purpose of this guide, we will assume we are modelling an HII region isobaric (constant pressure model) with $\log(P/k)=6.0$ with a metallicity $12+\log(O/H)=8.53$ and an ionisation parameter $\log(Q)=8.0$.

The following text is a commented sequence of typical inputs. Use this as an exercise. Enter one entry at a time, ignoring the ":" and subsequent text. In most cases, MAPPINGS ignores any string content after the required response. At each entry, the next options will be presented. You can explore these later, but this input file should suit most

simple models. You can edit the file and cut/paste the whole file at the prompt when you are happy with the settings.

Sample HII region single model input file for MAPPINGS

```
yes      : change abundance
abund/GC16Grid/GC_ZO_M0530.abn
no       : no more changes
no       : no offsets
no       : Use Kappa electron distributions
yes      : include dust
yes      : change depletions
abund/unified_depletion/Depln_Fe_1.50.txt
no       : no more changes
no       : allow grain destruction
M        : MRN distribution
yes      : Include PAH molecules?
0.3      : fraction of Carbon Dust Depletion in PAHs
Q        : PAH switch on QHDH < Value
4e2      : PAH switch on Value
no       : graphite grains to be cospatial with PAHs
no       : Evaluate dust temperatures and IR flux?
P6       : the main Mappings model to use
D        : Default ionisation values
H        : Input spectral energy distribution data (usually Starburst99)
Q/inputs/cont_a05t23isp_vm802.spectrum
9        : Age of the HII region. Age = (n-1)*0.5 Myr
X        : eXit with current source
N        : Include cosmic ray heating
S        : Spherical Geometry. (For Plane parallel, 'P', different options)
L        : Source by Luminosity
T        : Total or Ionising Luminosity
40       : bolometric source luminosity (log erg/s)
B        : isoBaric, (const pressure)
6.0      : Pressure regime (p/k, <10 as log)
4        : log(Initial temperature)
1        : filling factor (0<f<=1)
q        : Give initial radius in terms of distance or Q(N) (d/q) ***nb old
options
8.0      : Q at inner radius (< 100 as log)
y        : Volume integration over the whole sphere? (y/n)
E        : Equilibrium ionization balance.
0.02     : Step value of the photon absorption fraction *****
A        : Ionisation bounded, 99% neutral *****
A        : Standard output
SPH_zM0530_Q80_Pk6_cont_a05_t23_step9_M5118 : ID string
X        : end model
```

The model starts running as soon as the ID string is entered and the interim values for a series of physical parameters are displayed on the terminal screen as the computation proceeds. The final option, after the model finishes, asks if you want to run another model or exit. Enter X to terminate.

Depending on the output requested (here, "basic"), several files will be generated containing the model outputs. The most useful is the "specNNNN.csv" file which lists the line fluxes for all the lines above the threshold set in the MAPPINGS source file. In a text editor, inspect all the .csv and .ph6 files to see what they contain.

To run the whole model as a single exercise, you can cut and paste model text as a single block at the initial command line prompt after MAPPINGS starts, but you will need to trim the text following text input (e.g., the ID string). The response for "yes" can accept "y" or "Y" or "yes" or "YES", and similarly for "no". Upper or lower case letters are treated the same for simple responses. You can edit and save blocks like this as text files for the inputs for single models.

Grid models

It is straight forward to create script files to automate running models (shell scripts and/or awk files). This allows one to build a grid of models for a range of input parameters (e.g., metallicity, $\log(Q)$). In the HIIGrid306 directory there is a file named `rungrid.sh`. Examine this in a text editor to see how a shell script can be constructed. It can be edited in the same way the single model files to change model type (plane parallel or spherical), pressure, etc. This shell script creates a grid of models for metallicity (relative to standard or "solar") $Z = 0.05$ to $Z=3$, and ionisation parameter $\log(Q) = 6.50$ to 8.50 , values that span the range of the parameters likely to be encountered in an HII region.

Outputs: Apart from the model files (each metallicity set is copied at the end of the model run to its own $\log(Q)$ directory), the script (and associated awk scripts) creates two csv files with an array of line ratio values for each $Z/\log(Q)$ combination. One file orders results by Z , the other by $\log(Q)$, ready for plotting a grid of connected points for any two chosen strong line ratios.

We use an OSX application, Graf, to plot the grids, to which it is exceptionally well adapted, but you can plot the grids in any plotting program of your choice.

To run a grid model, go to the directory that includes `rungrid.sh` and execute it:
`./rungrid.sh` It will prompt you for the input parameters it needs, These are the number of core processors to use (5 is a good choice on a 4-core MacBook, for the 9 model sets to be run) and a name for the model. We are developing a standard for model names - stay tuned.

When you have run a grid model, examine the csv grid files. You will see columns with various physical parameters (e.g., metallicity, gas phase and total), and **log** line flux ratios for the different strong line combinations.

Note our convention for naming the ratio column headings: a "+" sign means the fluxes of both lines in a doublet are added. Entries without a flux refer to the brighter line in a doublet, or a single line. Thus [OIII] means the flux of the OIII line at 5007Å, while the "+" sign in [OIII]+ means flux ([OIII] 5007 + [OIII] 4959).

MAPPINGS input parameters

MAPPINGS' modelling capabilities extend far beyond the HII region models described here, so the following description of the model inputs is a subset, sufficient for this guide. Note that if you run a model one prompt at a time, MAPPINGS tells you all the possible options for the context. The prompts for an HII region model are as follows.

Change abundance: This allows you to select from the MAPPINGS libraries the abundances of elements to be used in the model. Of the 30 elements in MAPPINGS range, we mainly use the 16 elements of interest and importance in HII regions. This can be extended for more complex models but is seldom necessary.

Offsets: You can change abundance offsets for particular elements without changing the main abundance table. Normally ignore.

Use kappa distribution electrons: Nicholls et al. ([ADS 2012ApJ...752..148N](#)) suggested that under some circumstances, the electron energies may not be in equilibrium, but follow a kappa distribution. MAPPINGS can calculate fluxes for non-equilibrium electron energies. If you wish to model a normal Maxwell-Boltzmann equilibrium, respond "no" to this input. If "yes" you can specify the value of the kappa index ($2 < \text{kappa} < 1000$).

Dust and Change depletions: Normally include dust. Depletions are problematic as they are likely to vary from object to object. We have derived a set of scaled depletion files (see Nicholls, Sutherland et al, 2017, [ADS 2017MNRAS.466.4403N](#)) based on ISM measurements and analysis by Jenkins. They are based on the level of Fe depletion. We have found that a $\log(\text{Fe depletion})$ of -1.50 gives a useful match for observations of HII regions and PNe.

Dust destruction, grain distribution, PAHs and carbon in PAHs, etc: Use the default values shown in the model list above. Dust physics involves considerable degeneracies, and it is best to use the default values. These values generate results that match observations. If you enter responses to MAPPINGS prompts by hand, you will see the other options available and some explanations. Normally use the defaults, unless you wish to explore the effect of varying particular parameters, and know what you are doing!

MAPPINGS model to use: For HII regions, the choice should be P6, the most recent and most efficient code. Other P series are included for compatibility with older versions of MAPPINGS. You can model single atoms at fixed conditions (e.g., temperature, density) and if you choose MM, for example, the subsequent menus vary from what is described here. For this work, use P6.

Ionisation balance: normally use default

Radiation field: This refers to the source of the ionising radiation for the excitation. The choice allows you to select, *inter alia*:

- a black body,
- a single star for which you have the spectrum, from a variety of stellar atmosphere model sources,
- a synthesised cluster population spectrum calculated by, e.g., Starburst99, for which a pre-calculated library of spectra are included in the MAPPINGS data files, and
- An AGN spectrum.

As before, if you enter responses at the prompts manually, MAPPINGS shows all the options. For this work, choose H, which then requires choosing a spectrum file from the library.

Spectrum to use (file name): This requires selecting the appropriate spectrum file. In this exercise, we use Starburst99 files. The SB99 library includes precomputed spectra for a

range of cluster ages (0 to 10Myr) and for continuous star formation or coeval evolution at a single age. The library has been calculated for Salpeter and Kroupa initial mass functions (IMF), stellar atmosphere models, and stellar evolutionary track models. For example, the file `cont_a05t23isp_vm802.spectrum` has been calculated using continuous star formation, a Pauldrach WM-Basic atmosphere model, a Geneva evolutionary track and Salpeter IMF. The particular file matches the metallicities of the atmospheres and tracks (but see "Dark Secrets" below).

Model time (star or cluster age): Each spectrum file includes the models for different ages (0 to 10Myr in 0.5Myr steps) and this input selects which age to use. HII regions are brightest between 4 and 5Myr, options 9,10,11, so these values are the most useful.

Enter X at this point to tell MAPPINGS to continue.

Cosmic ray heating: normally default to "No".

Spherical or plane parallel geometry and Radiative transfer mode: MAPPINGS can calculate two geometries for HII regions. Spherical geometry corresponds to a Strömgren sphere. This is observed in small single star HII regions but is not the norm. Plane Parallel is the alternative, and it is usual to calculate a two sided model (radiation from the far side coming back through the nebula).

Density structure: this option allows a choice of constant density (isochoric, C), constant pressure (isobaric, B), or a functional form. For most HII regions, the sound crossing time is about the same as the age of the observed ionising cluster, allowing pressure to equilibrate, so constant pressure is the sensible choice. However, there are many combinations of metallicity and ionisation parameter at constant pressure that do have nearly constant density (see Kewley, Nicholls, Sutherland, et al., 2017, ARAA, in press). For this exercise, we use a constant pressure (isobaric) model.

Total pressure: This specifies $\log(P/k)$ where k is the Boltzmann constant. A value between 4 and 8 is plausible, and the best choice for observed HII regions is usually 5 or 6.

Initial temperature estimate: Electron temperature is a free parameter, not an input parameter for the model, but we need to tell MAPPINGS approximately where to start. It rapidly iterates to a physical value from the input parameters (P, Q, Z). $T_e = 10^4$ is the usual input value (4 as a log) for an HII region.

Filling factor: Normally use 1

Ionising flux at inner edge: Here we want to specify the ionisation parameter, $\log(Q)$, at the inner edge (point closest to the ionising source/cluster) so specify Q.

QHDN at inner edge: This specifies the value of the ionisation parameter at the inner edge. The usual range is $6.50 < \log(Q) < 8.50$.

Geometrical dilution factor: Another adjustment factor, normally use 0.5.

Ionisation balance: Normally enter E for equilibrium balance.

Step photon absorption fraction: This is the integration step size. For the P6 model, 0.02 is a good choice. A smaller value implies a longer computation time but may be required if the model is unstable at the larger step. Normally not a problem.

Model ending: Do we run out of ionising radiation first, or matter or optical depth etc? The simplest model is radiation bounded (A) and this is a good approximation to most nebulae.

Output settings: This is where we choose what output files are generated. We can select particular elements or particular ions to present, as a function of nebular depth, in addition to the standard output. For this exercise we select the standard output (A). Other options require additional inputs (element, ion, etc.).

Name/code for model run: We don't have a standard for this yet, but something that summarises the input parameters is useful. For testing purposes, "test" is fine.

When the run name has been entered, the model commences. If you are running a single model, the progress of the computation cascades through the terminal screen. The final input asks whether a new model should be run or the modelling ended. Enter X unless you want to run another model with different parameters.

And that describes the single model inputs. Next we'll take a quick look at the grid script, `rungrid.sh`. Many of the same inputs are used in the script as in the single model.

Grid script structure and inputs

The script file `rungrid.sh` (with associated awk scripts) is located in the HIIGrid306 directory. Open it in a text editor to see its structure and embedded inputs. It will be apparent that the first 150 lines relate closely to the single model inputs. After that the multi-model loops and scripts are set up, but the same parameters apply. More informative comments are provided in the script than are available in the single model input described above.

In particular, lines 81 to 97 set up a range of abundance files for five different metallicities. We are using by preference the Galactic Concordance abundance files (and scaling) described in Nicholls, Sutherland et al. ([ADS 2017MNRAS.466.4403N](#)). We use these in place of the so called "solar" abundances, which have varied greatly over the past 30 years. For example, standard metallicity (equivalent to 1.0 "solar") corresponds to $12 + \log(\text{O}/\text{H}) = 8.76$. However, equivalent files for standard (Asplund et al., 2009) solar values are also available in the `/abund/AGSS2009` directory. Older "solar" abundance files are also available in the `/abund` directory.

The relative abundance scaling of different elements in the Galactic Concordance set is based on the evidence in old stars. It is possible to express the abundances in terms of either the Fe scale or the oxygen scale. They involve the same physics, but the oxygen abundance files span a somewhat greater range of metallicities. The need for non-uniform scaling at different metallicities is described in Nicholls, Sutherland et al., ([ADS 2017MNRAS.466.4403N](#)). You can use the abundance files listed in the `rungrid.sh` file as inputs for single models or choose other standards from the `/abund` sub-directory in the `/lab` directory.

Other points to note

If you terminate a running script mid run using control-C, you will need to manually clean up the partly completed files in your working directory. If you are running multiple processes (e.g., if you run more than one process at a time), you will need to kill the zombie map51 processes. This can be done from the terminal, or, more easily followed, in the OSX Application Monitor app.

Dark secrets

As with all branches of astrophysics, those familiar with the assumptions that go into a model know that there are approximations involved, missing and unreliable data, and in some cases, Rumsfeldian "unknown unknowns". MAPPINGS has evolved over the years to incorporate the most efficient computation methods, the best atomic data available, and all the "known physics". The biggest problem we face is that, to determine the ionising spectrum of a star cluster, the synthetic spectrum depends on having extensive stellar atmosphere sets, and reliable interpolation between models (with a closely spaced model grid). The available O-star grids are sparse. We also need stellar evolutionary tracks calculated on the same basis (relative metallicity values) as the atmospheres. No such compatible sets exist at present, and we need to choose the most compatible combinations of atmospheres and tracks.

We also know that stellar atmosphere model fluxes, particularly for the massive O-stars that dominate the cluster ionising spectrum, can differ by an order of magnitude between the different atmosphere models at energies responsible for ionising O^+ to O^{++} , a critical value in determining nebular ionisation balances.

We are in the process of building a new grid of O-star stellar atmosphere models, TOSCA, using the CMFGEN code (Hillier, [ADS 2012IAUS..282..229H](#)). This is a huge undertaking and may not be complete for a while. Likewise, we are using the MIST code (Choi et al., [ADS 2016ApJ...823..102C](#)) to develop compatible evolutionary tracks. When both of these are complete we will use the SLUG2 stochastic stellar population spectral synthesis code (da Silva et al., [ADS 2012ApJ...745..145D](#)) to generate more consistent ionising spectra to use in MAPPINGS calculations.

Atomic data are also a problem. Because of the radiative cascade processes involved in generating recombination emission lines, lack of information about the higher energy levels of ions such as O^+ and C^+ leads to incomplete information on these transitions.

Nitrogen also presents a problem. Because of the different ways nitrogen can be created in stars (core collapse supernovae, WN stars, AGB stars etc.) the nebular abundance of nitrogen can vary considerably from a "standard" level in different nebulae. MAPPINGS can set the abundance of nitrogen to the desired value, but this must be adjusted manually when running a model.

These difficulties are faced by *all* photoionisation models. Despite this, it is gratifying how well we are able to model nebular photoionisation physics. The models provide a useful guide to observed physical conditions, but should not be assumed to be "absolute truth". Nonetheless, MAPPINGS is a very useful tool for understanding observations.

The abundance files, what they mean and how to use them

In the folder `mappings_V-xxx/lab/abund` (among other places) you will find a collection of element abundance files with the `.abn` suffix. These are derived from several sources, including the Asplund et al., 2009 [ADS 2009ARA&A..47..481A](#) "solar standard". These are scaled in a simple manner: e.g., 50% standard (solar) oxygen corresponds to 50% standard (solar) iron, which we know to be incorrect. In previous versions of MAPPINGS, these were the abundance sets we used, but should not normally be used except close to standard (solar) metallicity or to re-run old models.

As a result of identifying from stellar spectra how elements scale relative to one another at lower metallicities, we now have the "Galactic Concordance" (GC) scale, and the GC standard set. The latter is intended to replace the diverse "solar" abundance sets that have been measured of the past few decades, and which differ from each other by disturbingly large amounts. The GC standard is based on an extension of the abundances measured by Nieva & Przybilla, 2012, [ADS 2012A&A...539A.143N](#) from 20 local Milky Way B-stars. See Nicholls, Sutherland et al, 2017, [ADS 2017MNRAS.466.4403N](#), for a detailed discussion. At lower metallicities, the abundances are scaled following the stellar data.

The GC abundance files are provided for convenient line ratio grid calculations in the folder `../lab/abund/GC16Grid` and as a set of conveniently scaled values in the folder `../lab/abund/GC2016`. The `.abn` files are all text files. They include detailed header information on the abundances of oxygen and iron.

A word on the file nomenclature for the GC abundance files is in order here. The "M" and "P" notations mean minus and plus. Thus, `GC_ZO_M1130` means the standard oxygen metallicity ($12+\log(\text{O}/\text{H})=8.76$ for the GC scale) minus 1.130. So `GC_ZO_M1130` refers to a set with oxygen metallicity (8.76-1.13), or 7.63. The `GC_ZFe` set are derived from the standard Fe abundance, $12+\log(\text{Fe}/\text{H})=7.52$ (not the standard O abundance). So `GC_ZFe_M1150` means the standard Fe metallicity minus 1.15: $12+\log(\text{Fe}/\text{H})=7.52-1.15$, or 6.37. Using the GC relative scaling procedures, this corresponds to an oxygen metallicity $12+\log(\text{O}/\text{H})=8.11$.

Take a look at the contents of the file `GC_Fe_M1150.abn` (it's a text file) and you will see all this in the header comments. Open the other `.abn` files to see the oxygen and iron abundances each one corresponds to.

The `GC_ZFe` Grid files allow you to choose $12+\log(\text{O}/\text{H}) = 8.110, 8.485, 8.635, 8.835$ and 8.985 (`GC_ZFe_M1150, ...M0550, ...M0250, ...P0150, ...P0450`)

The `GC_ZO` files allow you to choose $12+\log(\text{O}/\text{H})= 7.630, 8.230, 8.530, 8.930$, and 9.230 (`GC_ZO_M1130, ...M0530, ...M0230, ...P0170, ...P0470`)

The `GC_ZO` files span a wider range of oxygen metallicities but using the `GC_ZFe` files allows you to choose a slightly different group of oxygen abundances. The physics and scaling procedures are the same, they just use different reference standard elements.

If you wish to run a single model at a specific metallicity, check the available metallicities in the `.abn` file headers in `GC16Grid` and `GC2016` folders to find the value nearest to the metallicity you want to model. You will also need to select the stellar atmosphere + track `.spectrum` file from the folder `../lab/HIIGrid306/Q/inputs/` that matches the `.abn` file as closely as possible.

The `.spectrum` files

MAPPINGS expects an excitation spectrum as input to ionise the ISM. For HII regions, the excitation source is usually a hot young star cluster. Shock excited spectra are also possible in HII regions, but modelling shocks is complex and outside the scope of this introductory guide.

MAPPINGS takes its excitation spectra in the form of `.spectrum` files, which can be found in the `../lab/HIIGrid306/Q/inputs/` folder. For typical star clusters we need to calculate the synthesised spectrum from a simulated young star cluster. This requires a value of the initial metallicity, inputs from an array of stellar spectra, a set of stellar evolutionary tracks, an Initial Mass Function (e.g., the Salpeter IMF), the age of the cluster and whether the cluster stars formed immediately and have all evolved since then, or are forming continuously. At present the stellar population synthesis program Starburst99 [ADS 1999ApJS..123....3L](#) is used to pre-calculate sets of spectra to use as inputs to MAPPINGS. As discussed above under "Dark Secrets", the stellar synthesised spectra are sub-optimal, but will improve as better stellar atmospheres, compatible evolutionary tracks and more sophisticated stellar spectrum synthesis applications are available.

At present, for a chosen model metallicity, one needs to select the appropriate spectrum file to run a sensible model. The following is a description of the file naming process we use to identify the right `.spectrum` file to choose to model a particular metallicity.

File names are of the form `cont_a03t21isp_vm802.spectrum`. The first section is either `cont` or `coeval`: the former refers to continuous star formation, the latter star formation at year zero only. The second section is typically `a03` or `a05`. This specifies the stellar atmosphere models used by Starburst99. `a05` has perhaps the best UV spectra and is based on the Pauldrach WMbasic code [ADS 2001A&A...375..161P](#). Data using CMFGEN will be available later and will cover a much finer temperature and metallicity grid. The next segment of the file name is `t21` in the example. This refers to the stellar evolutionary tracks, from the Geneva track models. The range of options in this case is `t21` through `t25`. These refer to metallicities for the grid metallicities 0.05 (`t21`), 0.2 (`t22`), 0.4 (`t23`), 1.0 (`t24`) and 2.0 (`t25`). It is this name segment that identifies the metallicity to be used in the (single) model. The next name segment is the IMF: `isp` for Salpeter, `ikr` for Kroupa. In practice, at present, the IMF choice makes little difference to the output due to the sparse O-star grid. The final `vm802` refers to the version of Starburst99 used. Each `.spectrum` file includes a sequence of spectra for different cluster ages, which is selected in the cluster age input when running a MAPPINGS model. Inspect a `.spectrum` file with a text editor to see the structure.

Using MAPPINGS to generate single-atom fluxes

Mappings can also be used to calculate emission line fluxes for single atoms and ions (for example, O⁺⁺). This is useful if you wish to calculate *atomic equivalent temperatures* for the T_e or "Direct method" to estimate electron temperatures from observed nebular fluxes. This is the electron temperature of a dust free, isolated assembly of ions at a uniform electron density and a single electron temperature that generates the same emission line flux as the observed nebular data. It can be useful when a nebula is assumed to be isothermal, and is widely used in nebular data analysis.

To run a single atom calculation we start MAPPINGS as normal, but respond with "n" to the prompts for abundances, abundance offsets, kappa, and dust. At the Model choice prompt, we enter "mm" (or "MM") to select Multi-level Ion Emissivity. This presents two options, depending on whether we want fluxes at a fixed electron temperature over a range of densities, or a fixed density over a range of electron temperatures. The more useful of these is usually the second, selected using the "B" input. This is because nebulae are approximately isobaric (constant pressure) but not isochoric (constant density) or isothermal. It is useful to calculate how the fluxes vary with temperature, but the results are the same for any temperature and density pair. We will illustrate the second case. Enter "b" (or "B").

The next option is to select the sort of output we want. The first leads us to a choice of collisionally excited ion, the second a many-level Fe model and the third, recombination line options. Here we'll select the first, enter "a". The ion list is presented, for which collisionally excited emission line fluxes can be calculated. We'll choose "12", for O⁺⁺. MAPPINGS shows the ion energy levels to be used, and asks for the density. Enter "2" for a density of 100 cm⁻³. Then we need to specify the minimum T_e (linear or log), the temperature increment delta (log only), and the number of temperature increments. Enter "5000, 0.05, 20", which corresponds to 21 temperature values starting at 5000K and ending at 50000K. Finally, give the run a name, and MAPPINGS creates a file named `ionxxxx.csv` with the results, where xxxx is a sequential number. MAPPINGS does not overwrite older files of the same type, but uses the next sequential value of xxxx. The final block is the requisite output flux data, listed for each emission line.

In summary, the script would look like this:

```
n      : no abundance change
n      : no offset change
n      : no kappa
n      : no dust
mm     : Multi-Level Ion Emissivity
b      : Fixed density, vary temperature
a      : Collisional, Multi-level Ions
12     : O III
2      : density 100 cm-3
5000, 0.05, 20 : start at 5000K, 0.05 log(T) increments, 20 steps
my_run_name : an identifier
```

How does MAPPINGS compare with CLOUDY?

Compared to CLOUDY, MAPPINGS is a small and relatively compact application, that runs quickly on a standard laptop. It is explicitly aimed at calculating nebular emission spectra. It handles some things that CLOUDY does not, for example shock emission spectra and kappa electron energy calculations. At least at present, CLOUDY does not scale nebular abundances to take into account relative abundance variations due to the evolution of elements. MAPPINGS does not calculate hydrogen recombination spectra from first principles, as CLOUDY does, but uses a table lookup method. Both codes are subject to the problems with atomic data and stellar atmosphere models. Neither code calculates nebular models with an arbitrary geometry (as MOCASSIN does).

Who wrote MAPPINGS?

MAPPINGS has been built by several authors over the years. Initially by Mike Dopita, then Luc Binette, Ralph Sutherland and Brent Groves. The most recent two versions (4, 5) have been built by Ralph Sutherland (with some input from David Nicholls and Mike Dopita) and are a major revision of earlier versions.

Contact information for further help

David Nicholls (general help): david.nicholls@anu.edu.au mob: 0458 901 888

Ralph Sutherland (*el supremo* inquiries only): ralph.sutherland@anu.edu.au

Appendix: the MAPPINGS Readme file

```
# MAPPINGS V v5.1.13
```

```
`https://mappings.anu.edu.au/code`  
`https://bitbucket.org/RalphSutherland/mappings`
```

```
#### Contact:
```

```
Ralph.Sutherland@anu.edu.au
```

```
## Quick Start Compiling and Running:
```

```
> cd mappings/src  
> make build`
```

```
[wait a while....]
```

```
> cd ../lab  
> ./map51
```

```
### Installing so map51 can run anywhere:
```

```
> cd mappings/src  
> make build  
> sudo make install
```

This copies map51 into /usr/local/bin so it can be run from any location by all users of the computer. It also creates copies of the essential lab directories; data, atmos, and abund, into /usr/local/share.

```
#### When running map51, you can run locally in lab/ (note ./map51)
```

```
> cd ../lab  
> ./map51
```

and it will find the local map.prefs, data, abund etc old scripts will work as before.

```
#### With the `/local` installed option you can use any directory  
(note plain map51) ie:
```

```
> cd  
> mkdir test  
> cd test  
> map51
```

If mappings finds a local data file it will use those, otherwise it will fallback to use the shared copy, so you can still maintain a custom set of data and abundances etc or use the standard shared set.

```
## Uninstalling:
```

To remove the shared installation simply


```
> sudo make uninstall
```

To remove the local copy simply delete the `mappings/` directory

The MAPPINGS directory structure:

```
#### mappings/
```

```
  lab/
```

```
    data/
```

```
    abund/
```

```
  src/
```

```
    mastercode/
```

```
    includes/
```

```
    workcode/
```

```
#### Optional:
```

```
  lab/atmos/
```

```
  lab/scripts/
```

```
#### Shared: (optional with sudo make install)
```

```
  /usr/local/bin/
```

```
  /usr/local/share/mappings/
```

```
    data/
```

```
    abund/
```

```
    atmos/
```

Key Directories and Files:

* `lab/` This is where the executable is made and run. The runtime files such

as map.prefs are here and the essential data directory with the runtime data files.

* `map51`: The executable.

* `map.prefs` Essential startup data - must be present.

* `mapStd.prefs` A standard 16 atom startup in case map.prefs is lost for any reason, can be copied and renamed map.prefs if needed

* `mapFull.prefs` A full 30 atom startup in case map.prefs is lost for any reason, can be copied and renamed map.prefs if needed

* `data/`: Contains the atomic data for MAPPINGS V. Read at runtime in mapinit.f. Essential and must be present and complete.

* `abund/`: A set of useful abundance settings that can be read interactively during a run. Optional.

* `atmos/`: A set of useful radiation source files and stellar atmosphere models. Optional.

* `scripts/`: A collection of (mostly) useful of UNIX shell and MAPPINGS scripts

* `src/`

Contains the code area and the Makefile(s). There are three makefiles, one

for a setup with fortran installed, Makefile.for, and one to use if you are running

f2c and a C compiler, Makefile.f2c. If using f2c, rename the Makefile.f2c to

simply Makefile and use that. If using FORTRAN, rename Makefile.for to Makefile.

All the useful parameters are near the top of the makefile.

* `Makefile`. This makefile controls all the building of MAPPINGS. It takes one argument to control the operation:

MAPPINGS V v5.1.x make options:

'make help' to see this menu

'make build' to create executable from scratch and clean

'sudo make install' to install the built executable into /

usr/local

'make prepare' to copy over '*.f' and '*.inc' files

'make compile' to create executable

'make backup' to backup '*.f' and '*.inc' files

'make clean' to backup '*.f' and '*.inc', remove '*.o'

'make listing' to create listing of code

* `mastercode/` Contains a master copy of the code - only accessed by the Makefile.

* `workcode/` The Makefile copies the .f files here. These can be edited etc by

the user. The Makefile compiles these files to make the executable.

* `f2c.zip`: A zip archive of `f2c`, requires only a `C` compiler to make, `gcc` by

default. MAPPINGS can be built from scratch by building `f2c`, installing it in

`/usr/local` and then use the `f2c` Makefile to build MAPPINGS.